

Symmetry and *symmetry breaking* are the organising principles for phases of matter. Recently, the discovery of new *quantum phases of matter* has challenged this paradigm. These phases are characterised by the *hidden* symmetry structures emerging in the *entanglement patterns* of the ground state wave function. Consequently, studying them requires an efficient description of the wave function that lays bare these entanglement structures. In turn, the study of these hidden symmetries naturally leads to *generalised* notions of symmetry no longer described by groups and their representations, but rather by *(higher-)fusion categories*.

In this dissertation, generalised symmetries present in strongly correlated quantum many-body systems are studied using *tensor networks*. Within this framework, both the local and non-local features of quantum many-body systems are described in terms of local tensors. The tensors directly capture the entanglement among the constituent quantum particles rather than the full wave function. The distinct ways in which these tensors are left invariant under a generalised symmetry are in one-to-one correspondence with distinct phases of matter protected by that symmetry.

The generalised symmetry operators themselves are also naturally represented as tensor networks that can be freely *pulled through* the ground state tensors. By exploiting these representations, the *gauging* of these symmetries and their associated *anomalies* are addressed in this thesis. In this context, gauging is a procedure relating theories that describe the same physics but have very different microscopic realisations and generally different symmetry structures. Such *dualities* are also represented as intertwining tensor networks that map between local ground state tensors. Anomalies are obstructions to this procedure. They strongly constrain the allowed phase diagrams, since they rule out trivial symmetric phases.

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An entanglement perspective on generalized symmetries, dualities and anomalies

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